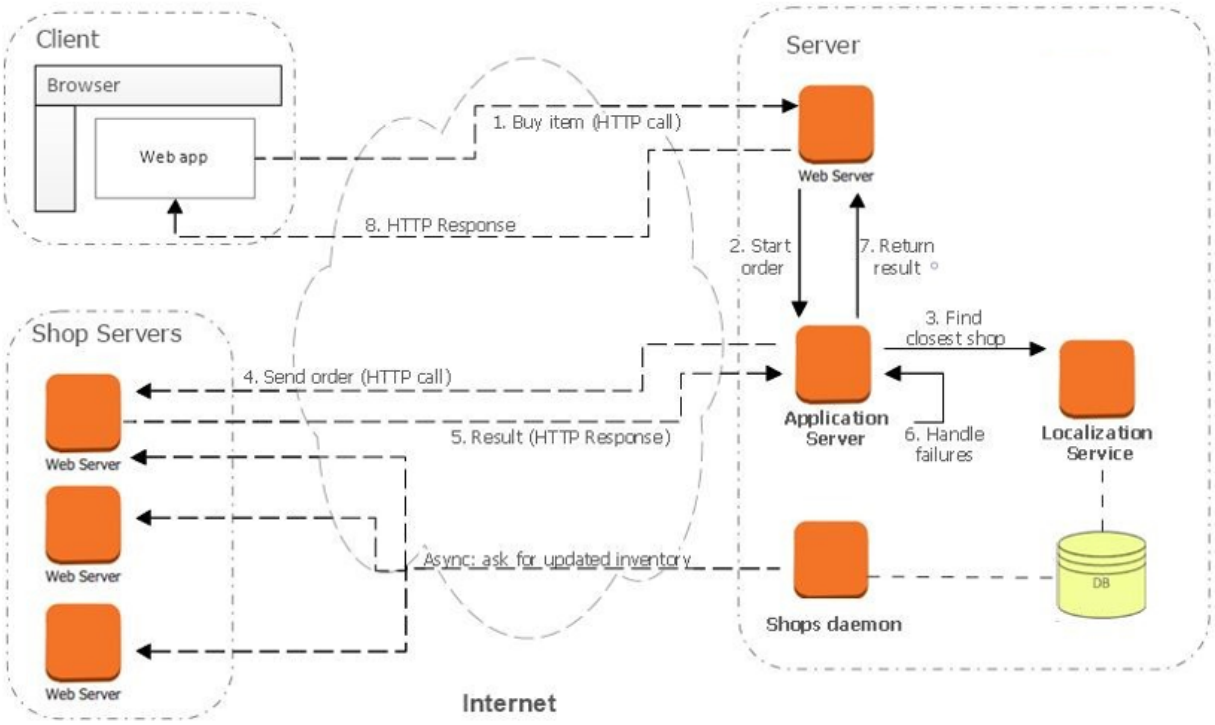


Figure 11.8 An even more advanced design, including a daemon that asynchronously polls shops' servers to sync the local DB to their inventories.

While we could think of workarounds for both issues, it is best to switch to a different approach: figure 11.8 shows an ever more complex architecture for our app, that includes a DB (could be something like Memcached, or a SQL DB, or a combination of both) and another service whose goal is just running (as a daemon, independently on the main app) and periodically ask the retailers' servers for an updated inventory; once a response is received, asynchronously, the service will update the local DB with the fresh values.

This also means that when we run the nearest neighbor search, we need to make sure that our in-memory copy of the inventory for the shops is up-to-date: here we will also have to compromise between performance and accuracy, as